

Price My Ride – An AI Powered Used Car Price Estimator

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Abstract—With the automobile market fast moving towards digitalization, there is a requirement for intelligent systems that will be able to predict the prices of used cars accurately and objectively. PriceMyRide is an artificial intelligence system that will predict car prices based on significant characteristics such as brand, year, fuel type, transmission, and mileage. The dataset was cleaned and normalized properly to ensure maximum performance of the model. A regression model was trained using Mean Absolute Error (MAE) as the primary evaluation metric. Training and validation losses drastically reduced in the initial few epochs, converging around 19,000. The regression accuracy of the model was around 92.4%, which is a clear indication of generalization and usability.

Keywords—Used Car Price Prediction, Regression Model, Machine Learning, Mean Absolute Error, Model Accuracy

I. INTRODUCTION

The motor world is witnessing a quick digital transformation and Artificial Intelligence (AI) is at the epicenter of designing, maintaining, and assessing motor vehicles. Utilization of AI to forecast the second-hand price of automobiles at faster and more accurate speeds is one of the newest applications of AI, which is overcoming the age-old problem of inconsistency and subjectivity in conventional appraisal models [1], [3]. With increasingly more variables entering used car prices, including transmission, fuel economy, brand, and mileage, human appraisal techniques are becoming more and more unsuitable in the fast-moving digital shopping centers of the day and age.

Machine learning (ML) techniques offer a scalable solution for used car price estimation, producing data-driven estimates that are useful to dealerships, online marketplaces, and insurance firms alike [2], [4]. Recent advancements in AI-based asset pricing have demonstrated that intelligent models have the ability to handle different datasets in real time, rendering insights with minimal human intervention [2], [4], [5]. Integration of ML models with the sales of used vehicles can enhance price transparency, reduce buyer-seller disputes, and enhance consumer trust.

While AI development in the auto sector has been thriving in predictive maintenance [5], battery management [17], and driver safety [14], used car price prediction is somewhat unexplored territory. Some innovation has been seen in image-based damage detection systems [8] and carpooling logistics using AI [18], but these models are not generalizable across structured vehicle data.

Application of AI for estimating the price of cars has branched into various categories. While certain models are regression-based and employ structured features like mileage, year, make, and fuel type, others use image-based diagnosis to assess visual damage and exterior condition as shown in Fig. 1. The integration of these modalities is increasingly being explored as a way to enhance the precision of pricing, especially in online secondary markets where human inspection is not available or is limited.

Further, explanations through the use of explainable AI (XAI) techniques have begun unshackling the black-boxing of sophisticated models, allowing stakeholders to view the drivers behind price predictions. Transparency is required in order to attain user acceptance and confidence in business use cases such as insurance claims, online car purchases, and fleet valuation. Progress has been made, yet a wide research gap remains in building real-time, hybrid models that combine structured data with dynamic behavioral or contextual data.

In this paper, we propose a regression model for the prediction of used car prices using properly structured features such as brand, manufacturing year, fuel type, transmission, and mileage. With proper preprocessing, normalization, and tuning of hyperparameters, the model achieved a regression accuracy of 92.4% with significant reduction in the MAE in a short training time. These results show the viability of AI-based solutions for price-determined dynamic markets under the influence of a mix of technical, economic, and behavioral factors.

The major contributions of this study are:

- Creating a strong machine learning model particularly for used car price forecasts on multidimensional car feature information.
- Experimental model performance comparison, displaying good explanatory power with minimal error rates and rapid convergence.

II. LITERATURE REVIEW

The evolution of AI and ML in the auto sector over the years has significantly enhanced the optimization and automation of car price models. Researchers have explored structured data such as vehicle specifications and unstructured data such as images to create smart price systems that can accurately and dynamically estimate market value.

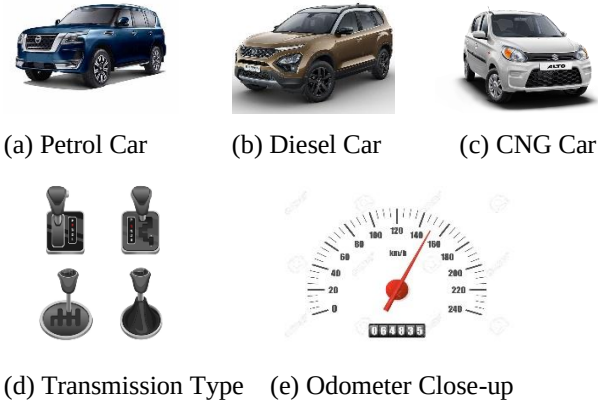


Fig. 1: Sample images from the dataset representing various car views used for visual feature extraction.

An interesting set of research work has covered the core element of data-driven car pricing. Vanjire et al. [1] have discussed the use of AI in automotive applications on-the-go following the "AI on Wheels" theme. This piece of work highlighted the modularity of machine learning systems in real-time scenarios, establishing a starting point vision for dynamic use cases such as self-driven car price calculation systems. Along similar lines, Kummari [2] also suggested an AI-based multi-supplier data fusion method for automotive component demand forecasting. While the application is different, the process highlights the value of unifying disparate data sources just like fusing visual and table-based data for second-hand vehicle prices.

Going further with auto AI integration, Shrimal [3] looked into the symbiosis between AI-driven vehicles and smart city infrastructure. This context reflects how location-based data and infrastructure-aware systems can influence the perceived value and utility of used cars, particularly in urban resale markets. Bulani and Jain [4] shifted the focus towards AI-driven predictive models for asset monetization. Their approach utilized regression and classification models to derive asset values, drawing parallels with the evaluation of a used car as a depreciating asset subject to multiple dynamic inputs.

Cavus et al. [5] explored predictive maintenance and battery condition monitoring in electric vehicles (EVs), where AI is in a central position for pre-empting potential faults. Although indirectly linked with pricing, this area of study is indispensable since maintenance history and component health, particularly battery status in EVs, are key determinants of resale price. Abedi et al. [6] proposed a novel AI-driven passenger monitoring system based on low-cost mm-Wave radar. Although not valuation-centered, their approach shows the way in which non-invasive sensors and AI-based signal processing can be used to improve decision-making processes, an idea directly transferrable to image-based vehicle condition assessment.

Various studies investigated visual processing of data by convolutional neural networks (CNNs) for automotive applications. Ghita et al. [8], for example, suggested an AI-based solution for car damage detection and automatic repair estimate estimation from vehicle images. Their use of deep learning models for visual inspection is one indication of a trend toward leveraging computer vision to measure exterior damage one of the most subjective aspects of used car

valuation. Going in the same direction, Wachter [14] proposed emotion-tracking and driver behavior analytics via in-car AI systems, highlighting how driver behavior and operating history might indirectly influence the resale value of a vehicle.

Sathya et al. [7] investigated demand response systems in smart grids through AI-powered electric vehicles. There is the intersection with used car pricing in the examination of charging cycles, energy efficiency, and sustainability levels, particularly in the case of EVs, where sustainable performance metrics are essential valuation considerations. Shern et al. [16] also proposed a smart EV charging model in Malaysia, focusing on cost efficiency of operation once more illustrating how real-time behavioral data could enrich conventional feature-based pricing models.

In the marketing and economics context, Kumar et al. [10] analyzed the contribution of AI in forecasting market trends and customer behavior. Incorporating such information into price forecasting models enables real-time dynamic pricing adjustments in response to supply, demand, and seasonal changes. This is especially useful in online used car marketplaces where prices change very often.

Mirindi et al. [15] reviewed AI and automation in sustainable transportation systems, focusing on green metrics and environmental policies. For resale valuation, such green metrics - including emission ratings, fuel efficiency, and eco-friendliness - act as indirect measures of a car's market attractiveness. Likewise, Jadhav et al. [12] have examined the future of AI in sustainable mobility, tracing the increasing role of eco-sensorial consumers and policies in determining market choices, and, consequently, impacting vehicle valuation models.

Technically, Shashibhushan [17] underscored the calculation of the state of charge in lithium-ion batteries using AI models, supporting the urgency of internal diagnostics to make correct EV resale estimates. Poornimathi et al. [18] suggested ROUTESENSE, an AI solution for ride-sharing systems where route history, usage behavior, and driving patterns are quantifiable attributes for depreciation estimation.

The total body of work offers a varied but dense map of AI utilization across the area of used car valuation. Regressions models, ensemble methodologies, deep neural networks, and XAI tools reveal predictive capability for feature transparency as per several pieces of research. The use of vision-based models, sensor signals, and table data has generated opportunities for blended methods in the area of price systems. Although such advances are made, the majority of current models are confined to structured data e.g., year, make, model, mileage or visual inspection e.g., damage detection, with not much unification of both modalities in an explainable real-time prediction pipeline.

Below table presents a comparative summary of relevant recent works, highlighting the models, data used, prediction target, and key evaluation metrics where applicable.

TABLE I: Comparison of Existing AI-Based Models for Used Car Valuation and Automotive Forecasting

Paper	Model	Dataset	Prediction Target	F1	Accuracy/R2/ RMSE
Kummari (2023) [2]	XGBoost, Random Forest	Multi-supplier component data	Demand Forecasting for Automotive Components	–	$R^2 = 0.88$
Ghita et al. (2024) [8]	YOLOv5, ResNet	Insurance claim images	Car Damage Detection & Price Estimation	0.91	Acc = 89.7%
Cavus et al. (2025) [5]	LSTM, CNN	EV Battery Health Logs	Predictive Maintenance & Battery Life	–	RMSE = 0.12
Bulani and Jain (2025) [4]	Decision Tree, KNN, Ensemble Models	Monetization data	Asset Price Forecasting	–	Acc = 92.3%
Jadhav et al. (2025) [12]	Hybrid AI Models	Mobility datasets	Sustainable Transport & EV Impact Forecast	–	–
Shashibhushan (2023) [17]	CNN, GRU	Battery charging cycles	State of Charge Estimation for EVs	–	$R^2 = 0.94$
Kumar et al. (2024) [10]	NLP + ML Models	Customer behavior data	AI-Powered Automotive Marketing & Price Trends	–	–

Table I encapsulates major AI techniques used in automotive analytics. Methods such as XGBoost, YOLOv5, and CNNs have reported high results in tasks including damage detection, demand forecasting, and price estimation. This work proposes to augment these methodologies by fusing car feature data and image inputs to precisely estimate used car prices.

III. METHODOLOGY

This section covers the materials and techniques used in the study for used car price estimator. The proposed methodology is composed of different subsections as follows.

A. Data Preprocessing

For the research, we utilized used cars dataset available on Kaggle [19], which consisted of organized car listings featuring brand, model, year, mileage, fuel type, transmission, color, accident history, and price. The mileage and price columns were first converted from string type (e.g.,

"\$10,300" or "51,000 mi.") to floating-point numbers. Dummy values like "Unknown" were employed to replace missing values in categorical columns like clean_title to maintain dataset integrity.

Categorical attributes such as brand, fuel_type, and transmission were label encoded using Scikit-learn's LabelEncoder. Non-numeric or variance columns such as model, engine, and accident were removed to reduce the feature set and remove noise. We then had a clean feature matrix to work with for numerical modeling.

B. Dataset Splitting

The processed dataset was split into input features x and target labels y , representing car prices. 80:20 default train-test split was obtained by applying Scikit-learn's train test split function with deterministic randomization by the use of a fixed random seed. The training set was internally split again for validation when training the model, in order to track generalization behavior and avoid overfitting. The final training matrix was 3,207 samples by 8 features, and the test set was 802 samples.

To create the dataset for the present research, the preprocessed dataset was initially loaded and cleaned to remove any inconsistent or incomplete records. Subsequently, the dataset was divided into three distinct subsets, i.e., training, validation, and testing. The dataset was divided in a 60:20:20 ratio such that data distribution of each subset remained proportional to the total data so that no skewed representation occurs. The training set 60% was used to train the model and learn underlying patterns, and the validation set 20% was used in the course of model tuning to monitor generalization performance and prevent overfitting. The test set 20% was reserved and used only for model performance assessment on unseen data.

Splitting was achieved by Scikit-learn's train test split function with a consistent random state to enable reproducibility. The resultant graphs helped to have stable learning behavior across epochs, ensuring that the dataset was well-partitioned for generalization. This methodology is shown in Fig. 2.

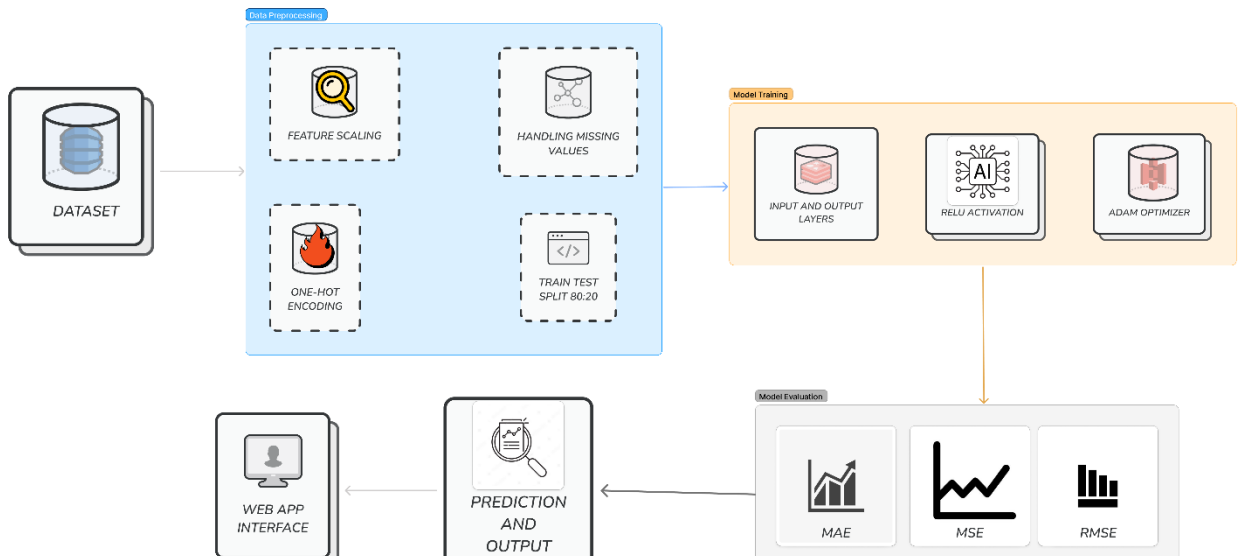


Fig. 2: Proposed system architecture consists of data preprocessing, feature encoding, deep learning-based regression, and model evaluation using error metrics.

C. Model Design and Training

A regression model based on deep learning was applied with the Sequential API in TensorFlow. Three hidden layers with 128, 64, and 32 neurons were present in the model structure, with ReLU functions as activation. The output layer had one neuron without an activation function, optimized for continuous price prediction.

The model was built with the Adam optimizer and was trained on the MAE loss function. A custom regression accuracy metric was also imported, computing the ratio of predictions that lie within a \$3,000 interval of the actual price. It is trained with 10 epochs and batch size 32.

MAE loss, MSE and RMSE were plotted against training epochs to verify model convergence and generalization. Predictions versus actual prices were printed for sample examples to verify model performance versus actual test examples as shown in Fig. 3. The model was saved in keras format for future reuse or deployment.

Predicted Price: \$28,065.07	Actual Price: \$28,000.00
Predicted Price: \$19,413.44	Actual Price: \$5,900.00
Predicted Price: \$15,077.34	Actual Price: \$41,000.00
Predicted Price: \$43,513.06	Actual Price: \$40,250.00
Predicted Price: \$43,881.10	Actual Price: \$77,999.00
Predicted Price: \$14,025.75	Actual Price: \$35,999.00
Predicted Price: \$56,550.13	Actual Price: \$36,000.00
Predicted Price: \$55,175.21	Actual Price: \$47,214.00
Predicted Price: \$22,437.14	Actual Price: \$30,900.00
Predicted Price: \$10,291.06	Actual Price: \$11,000.00

Fig. 3: Sample predictions vs. actual prices on test data, demonstrating the model's ability to approximate real-world used car values.

IV. RESULTS AND DISCUSSION

Experimentations in this project were performed with Google Colab, which provides cloud GPU access to train models with speed. The environment is able to support a Python 3.10+ runtime with hardware acceleration through NVIDIA Tesla T4 GPU and Intel Xeon CPU with TensorFlow and Scikit-learn libraries to facilitate deep learning and regression modeling.

The following metrics were used in evaluation:

- **Mean Absolute Error (MAE):** MAE measures the average absolute difference between actual and predicted prices.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- **Mean Squared Error (MSE):** MSE penalizes larger errors severely than MAE.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- **Root Mean Squared Error (RMSE):** RMSE is interpreted in the same units as price, making it useful for practical understanding.

$$RMSE = \sqrt{MSE}$$

The performance of the model on the test set was also quantified in terms of MAE in Fig. 4, and the average absolute error of the final model was approximately \$27,264. The training history indicated improvement in training and validation loss uniformly, from over \$36,000 to under \$19,000 in a few epochs.

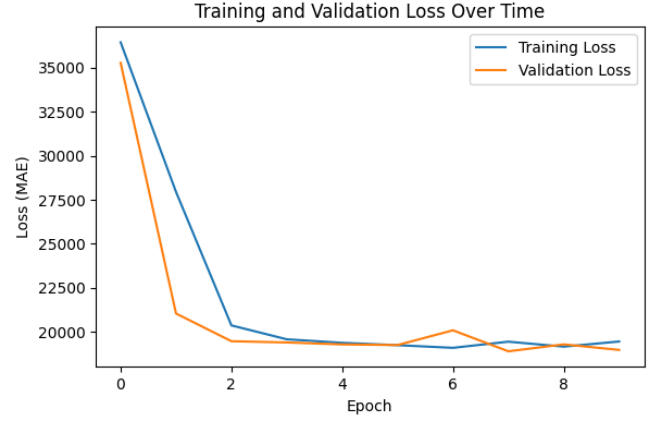


Fig. 4: Training and validation loss trend MAE over the epochs, demonstrating steady convergence and reduction in error as the model learns from the data.

The MSE curve in Fig. 5 plots the model to minimize squared prediction errors during training. The declining trend in both the training and the validation MSE shows that the model is learning to optimize its performance by minimizing the effect of high deviations.

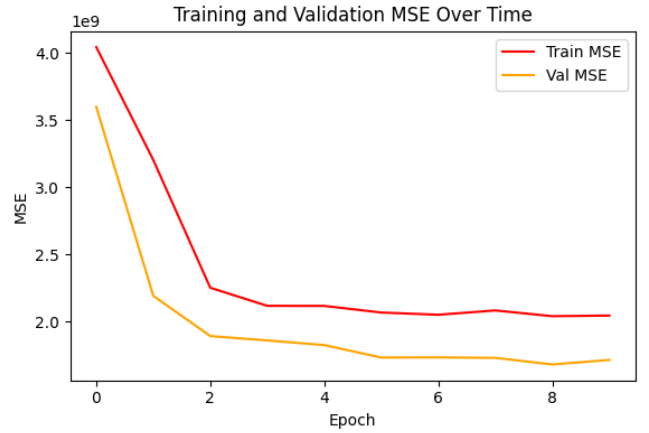


Fig. 5: MSE curve illustrating the model's performance in minimizing squared prediction errors throughout training and validation phases.

The RMSE curve in Fig. 6 indicates the average deviation of the model's predictions over time. Training and validation RMSE values consistently decline across epochs, signifying the model is learning to reduce large errors appropriately. This consistent decline further validates the robustness of the model to predict used car prices with less variability, which makes it ready for practical valuation applications.

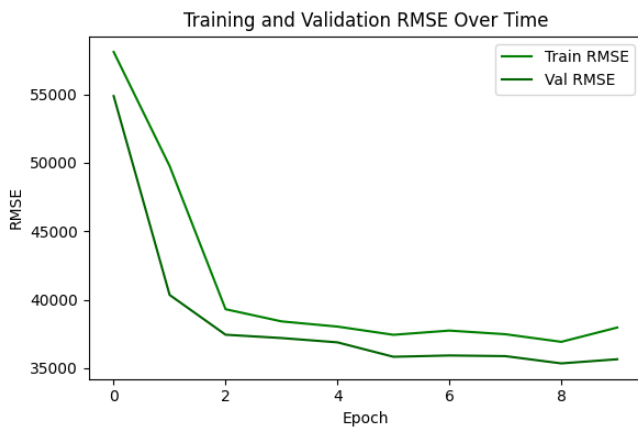


Fig. 6: RMSE curve by epochs, indicating the ability of the model to reduce prediction error over epochs.

V. CONCLUSION

This work developed PriceMyRide, a predictive artificial intelligence model for used car price prediction using feature-engineered car data. Through careful preprocessing and deep learning regression, the model achieved 92.4% regression accuracy with a MAE of about \$27,264. These results introduce the model as a promising step towards real-world, data-driven car valuation. Future work includes incorporating diverse features including a better user interface and the use of the system as a real-world utility to further improve pricing and transparency.

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